Component list

Bill Of Materials - PDF for Variant [Maximal Configuration] of Project [Nixie Clock Avito x6 - SMD.PrjPcb] (No PCB Document Selected)

Project:	**Nixie Clock Avito x6 - SMD.PrjPcb**
Variant:	**Maximal Configuration**
Author:	Dimasssius
Report Date:	09.04.2026

#	Designator	Comment	Description	Footprint	Quantity
1	ANT1	AN9520-245	Antenna, SMD, 2.1x9.5mm, 1.5dBi, Patch Antenna, 200MHz, 2.45GHz, 3W, Antennas ROHS	ANTC9520	1
2	B1	CR1025	Battery Clip Coin, SMD, CR1025, Cell	CR1025	1
3	C1	22uF	Capacitor, Ceramic (MLCC), SMD, 22uF, 6.3V, ±20%, X5R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	1
4	C2, C13, C16, C19, C26	1uF	Capacitor, Ceramic (MLCC), SMD, 1uF, 6.3V, ±10%, X5R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	5
5	C3	1uF	Capacitor, Ceramic (MLCC), SMD, 1uF, 250V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C1812_(M4532)	1
6	C4, C6	0.22uF	Capacitor, Ceramic (MLCC), SMD, 220nF, 6.3V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	2
7	C5, C9, C10, C11, C15, C21, C22, C23, C27	0.1uF	Capacitor, Ceramic (MLCC), SMD, 100nF, 6.3V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	9
8	C12, C14	6.8pF	Capacitor, Ceramic (MLCC), SMD, 6.8pF, 50V, ±5%, C0G, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	2
9	C17	10uF	Capacitor, Ceramic (MLCC), SMD, 10uF, 6.3V, ±20%, X5R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	1
10	C18	3pF	Capacitor, Ceramic (MLCC), SMD, 3pF, 50V, ±5%, C0G, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	1
11	C20	2.4pF	Capacitor, Ceramic (MLCC), SMD, 2.4pF, 50V, ±5%, C0G, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	1
12	C24, C29	100uF	Capacitor, Aluminum Electrolytic Capacitor, SMD, D5xL5.4mm, 100uF, 6.3V, ±20%, Aluminum Electrolytic Capacitors - SMD, ROHS	CAP_SMD_ELEC_ALUM_5MM_H5.4MM	2
13	C28	10nF	Capacitor, Ceramic (MLCC), SMD, 10nF, 50V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	1
14	D1	ES1JFL	Diode, SMD, SOD-123F, 600V, 1A, 22.55ns, 1.7V@1A, Independent Type,Fast Recovery / High Efficiency Diodes, ROHS	SODFL3719X135N	1
15	D2	IRM-V538	Infrared Remote Receiver (IRM), 38kHz, 8m, 1.2mA, 2.7V~5.5V, Infrared Remote Receivers (IRM), ROHS	IRM-V538	1
16	D4, D5, D6	LED0603	LED Indication - Discrete, 0603, 20mA, 225mcd, Colorless Transparent Lens, -20℃~+85℃, Positive Stick, 620nm~630nm, Red, 120°, 50mW, 2.3V, LED Indication - Discrete, ROHS	LED 0603_(1608)_0402A_(1005)	3
17	D7	SS12	Diode Schottky, SMD, SMA(DO-214AC), 20V, 1A, 500mV@1A, Schottky Diodes, ROHS	DO-214AC	1
18	IC1, IC2, IC3, IC4, IC5, IC6, IC7, IC8	TLP291	Photovoltaic Output Optoisolator, SOP-4-1.27mm, 80V, 3.75kV, 50mA, 300mV@2.4mA, 8mA, 1, 5V, 1.25V, DC Transistor, Photovoltaic Output Optoisolators, ROHS	SOIC127P700X230-4N	8
19	IC9	74HC595D	Shift Register, SOIC-16, Serial To Serial Or Parallel Shift Register, ROHS	SOIC127P600X175-16N	1
20	IC10	MAX668	AC-DC Controller and Regulator, MSOP-10, 100kHz~500kHz, 3V~28V, AC-DC Controllers and Regulators, ROHS	SOP50P490X110-10N	1
21	IC12	ATMEGA328P-AU	Microcontroller, TQFP-32(7x7), AVR, 20MHz, Microcontrollers (MCU/MPU/SOC), ROHS	QFP80P900X900X120-32N WITH SPI INTERFACE 2.54	1
22	IC13	W25Q64JVSSIQ	NOR FLASH, SOIC-8-208mil, 64Mbit, SPI, 133MHz, NOR FLASH, ROHS	SOIC8	1
23	IC14	ESP8266EX	Microcontroller, QFN-32-EP(5x5), 72.2Mbps, 2400MHz~2483.5MHz, WiFi 2.4GHz, RF Transceiver ICs, ROHS	QFN50P500X500X90-33N-D	1
24	IC16	AMS1117-3.3	Low Drop Out (LDO) Regulator, SOT-223, SOT-89-3, 72dB@(120Hz), 1A, Fixed 3.3V, Positive electrode, 16V, Voltage Regulators - Linear, Low Drop Out (LDO) Regulators, ROHS	SOT-89-3	1
25	IC17	RV-3028-C7	Real Time Clock, SMD3215-8P, I2C, Real Time Clocks, ROHS	SMD3215-8P	1
26	IC18	K155ID1	Decoder, DIP-16, 5V±5%, 25mA, 150ns, High Voltage Output, BCD to Decimal Decoders, RoHS	DIP16	1
27	J1, J6	Female Header 1x10-Pin	Female Header, 1x10-Pin, 10P, P=2.54mm, Direct Insert, H:8.5mm, W:2.54mm, D:3.2mm, Single Row, Black Brass, Top Square, Hole Plugin, 3A, 20mΩ, 1000V, -40℃~+105℃, Female Headers, ROHS	HEADER_FEMALE_10PIN_LOW	2
28	J2, J3, J4	Female Header 1x3-Pin	Female Header, 1x3-Pin, 3P, P=2.54mm, Direct Insert, H:8.5mm, W:2.54mm, D:3.2mm, Single Row, Black Brass, Top Square, Hole Plugin, 3A, 20mΩ, 1000V, -40℃~+105℃, Female Headers, ROHS	HEADER_FEMALE_3PIN_LOW	3
29	J5	USB_C	USB Connector, 5A, USB 3.1, 1 Horizontal Attachment, 16P, Female, -25℃~+85℃, Type-C, SMD, USB Connectors, ROHS	USB_C_DOUBLE	1
30	L1	2.2nH	Fixed Inductor, SMD, 0603, 2.2nH, 500mA, ±5%, 200mΩ, Fixed Inductors, RoHS	IND_SMD_0603_(M1608)	1
31	LS1	CMI-9605-0580T	Buzzer, D=9.6mm, 80dB, Active (Built-in Driver Circuit, Electromagnetic Type, 2.7kHz, Plugin, Buzzers, ROHS	CMI96050580T	1
32	Q1, Q2, Q4	MMBTA42	Transistor Bipolar (BJT), NPN, SOT-23, 300V, 225mW, 25@10mA, 10V, 500mA, NPN Bipolar (BJT), ROHS	SOT105P255X140-3N	3
33	Q3	AON7264E	MOSFET (N-Channel), DFN-8(3x3), 60V, 28A, 11W, 13.3mΩ@4.5V, 28A, 1.8V, 1 N-channel, MOSFETs, ROHS	DFN-8-EP(3x3)	1
34	R1, R15, R35, R36, R38	4.7kΩ	Resistor, Thick Film Resistor, SMD, 4.7kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	5
35	R2	470Ω	Resistor, Thick Film Resistor, SMD, 470Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
36	R3, R5, R6, R7, R18, R20, R21, R23, R29, R31, R33, R37	10kΩ	Resistor, Thick Film Resistor, SMD, 10kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	12
37	R4, R14	33kΩ	Resistor, Thick Film Resistor, SMD, 33kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	2
38	R8	8.2kΩ	Resistor, Thick Film Resistor, SMD, 8.2kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
39	R9	1.5MΩ	Resistor, Thick Film Resistor, SMD, 1.5MΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
40	R10	3kΩ	Resistor, Potentiometer, SMD, SMD-3P, 3.9x3.1mm, Potentiometer, 3kΩ, ±25%, 150mW, Potentiometers, Variable Resistors, ROHS	TC33X	1
41	R11	56kΩ	Resistor, Thick Film Resistor, SMD, 56kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
42	R12	499kΩ	Resistor, Thick Film Resistor, SMD, 499kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
43	R13	0.1Ω	Resistor, Thick Film Resistor, SMD, 0.1Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R1206_(M3216)	1
44	R16	100Ω	Resistor, Thick Film Resistor, SMD, 100Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
45	R17	3kΩ	Resistor, Thick Film Resistor, SMD, 3kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
46	R19	GL5528	Photoresistor, Through Hole, P=3mm, 10kΩ~20kΩ, 540nm, 1MΩ, Photoresistors, ROHS	Photoresistor	1
47	R22, R24	220Ω	Resistor, Thick Film Resistor, SMD, 220Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	2
48	R25, R26, R30, R32	1kΩ	Resistor, Thick Film Resistor, SMD, 1kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	4
49	R27	499Ω	Resistor, Thick Film Resistor, SMD, 499Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
50	R28	12kΩ	Resistor, Thick Film Resistor, SMD, 12kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
51	R34	5.1kΩ	Resistor, Thick Film Resistor, SMD, 5.1kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
52	S1, S2, S3, S4	TS-1009B	Switch Tactile Through Hole, 4.5x4.5mm, Round Button, 50mA, Bend-in 4.5mm, SPST, 12V, Plugin, Tactile Switches, ROHS	TS-1009B	4
53	S5	TSA016A2518C	Switch Tactile SMD, 3.35x2.5mm, Oval button, 50mA, Standing paste, 4.2mm ,180gf, 12V, SMD Tactile Switches, ROHS	TS20	1
54	S6	BME280	Development Board, LGA-8(2.5x2.5), Temperature and Humidity Sensor, ROHS	BME280_DEVBOARD	1
55	S7, S8, S9, S10	TS-1187A-B-A-B	Switch Tactile SMD, 5.1x5.1mm, Without bracket, 50mA, 100000 Times, 160gf, 12V, Round, SMD Tactile Switches, ROHS	TS-1187A-B-A-B	4
56	TR1	EE0504S	Power Transformer, SMD-5P, 1:10, 6.4x6.4mm, Power Transformers, ROHS	EE0504S	1
57	U1	CH340C	USB Converter, SOP-16, 2Mbps Transceiver, USB 2.0, USB Converters, ROHS	SOIC127P600X180-16N	1
58	Y2	CSTCE16M0V53-R0	Ceramic Resonators, SMD, SMD3213-3P, 3.2x1.3mm,16MHz, 15pF, 40Ω, Ceramic Resonators, ROHS	XTL_3_3213	1
59	Y3	XTM20026000FT00351005	Crystal Oscillator, SMD, SMD3225-4P, 3.2x2.5mm, 26MHz, 12pF, ±10ppm, ±20ppm, Crystals, ROHS	XTL_4_3225	1
Notes					**114**